FORMAL ENGINEERING & BUSINESS SPECIFICATIONS

Product Requirements Document (PRD)

Comprehensive architectural, operational, marketing, and legal requirements for launching NCAMart â€" The trusted e-commerce and local drop-and-pack tech platform in Nawabpur.

DOCUMENT VERSION
Version 1.3 (Final Approved Draft)

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Approved for Technical & Marketing Execution

PREPARED FOR
Executive Leadership & Marketing Management

PRIMARY AUTHOR
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Executive Briefing for Leadership & Marketing

This document defines NCAMart's strategic positioning to dominate Nawabpur, Sundarban Square, and Stadium Market electronics retail. Key focal points include: the **Two-Tier Trust Engine** (Authentic vs. Tested OEM Budget), **Physical VOID Security Seals** with online serial lookups, **Pre-Order Import Engine** with 10%â€"20% deposits, **Multi-Channel Marketing Automation**, and anti-fraud Cash-on-Delivery logistics.

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Phased timeline from Foundation (Phase 0) to MVP Launch (Phase 1) and Scale (Phase 2).

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Final confirmed specifications on deposits, payment gateways, branding & tech stack.

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Based on: idea.md, prd-analysis.md & requirement-validation.md

Change Log

Version	Date	Changes
1.0	—	Initial concept draft.
1.1	2026-10-14	Standardized deposit to 10–20%. Set strict no-replica policy for Budget Tier. Currency fixed to BDT only. Added initial operational sections.
1.2	2026-10-14	Incorporated Product & Business Analysis: Confirmed tiered deposit (10%/20%), SSLCommerz gateway, advance delivery fee on COD for first-time buyers, and resolved open questions.
1.3	2026-10-14	**Complete Technical Requirements Validation:** Integrated 6 missing operational flows (RTO, Pre-Order Forfeiture, Reverse Logistics Courier RMA, Showroom PIN Pickup, Supplier Re-quote, Distributor Warranty); 7 Edge Cases (Redis cart locks, in-transit cancellation locks, cloned seal detection, partial stockout, offline runner PWA); Full Error Handling architecture (SSLCommerz reconciliation cron, courier exponential backoff retry, thermal printer void protocol); Cryptographic Base32 Seal ID entropy; Granular 8-Role RBAC Matrix with runner PII masking; Multi-Channel Notification Matrix with BTRC quiet-hour compliance; and 5 Exception/Escalation Pathways.

1. Project Overview

1.1 Executive Summary

NCAMart is a specialized e-commerce and retail platform anchored in the Nawabpur Complex, directly adjacent to Sundarban Square, Stadium Market, and Patuatuli in Dhaka, Bangladesh. The regional tech and utility electronics market is plagued by pervasive counterfeits, misleading branding, grey-market speculation, and a refusal by merchants to honor warranties.

NCAMart solves this trust deficit through a core value proposition of **Radical Transparency**:

- Every catalog SKU is explicitly segmented into **100% Authentic** or `Budget Smart (Tested OEM)`, backed by visual badges and side-by-side spec comparisons. No counterfeit branding or replicas are permitted.
- An **In-House QC & Serial Verification Engine**, reinforced by physical **Destructive Tamper-Evident VOID Seals** and cryptographically randomized Seal IDs queried via `/verify`.
- An **On-Demand Pre-Order System** for specialized engineering tools and high-end smart gadgets, requiring a tiered **10%–20% advance deposit**, eliminating dead-stock risk.
- Fast **Drop-and-Pack** local fulfillment, utilizing hyper-local runner proximity to adjacent wholesale markets to ship orders same-day or next-day without warehouse inventory carrying costs.

1.2 Core Business Objectives

1. **Eliminate Customer Mistrust:** Establish an unshakeable reputation for honesty by removing ambiguity between genuine branded gear and budget alternatives through clear badges, upfront specs, and online serial verification.
2. **Zero Dead-Stock on Niche Tech:** Eliminate working capital lock-up by securing verified customer deposits before importing high-ticket engineering gear.
3. **Agile Local Fulfillment:** Ship standard accessories within 24 hours in Dhaka by dispatching runners to wholesale stalls on-demand.

1.3 Success Metrics (KPIs)

Metric	Phase 1 Target (first 6 months)	Purpose & Measurement
Website Conversion Rate	≥ 1.5%	Tracked separately for Authentic vs. Budget Smart tiers.
Return / Replacement Rate — Authentic	≤ 2%	Validates official distributor supply chain integrity.
Return / Replacement Rate — Budget Smart	≤ 5%	Monitored per vendor to eliminate low-grade wholesale stalls.
Verification Engagement Rate	≥ 25% of delivered orders	Measured via QR scan and Seal ID lookups on `/verify`.
Pre-Order Fulfilment Rate	≥ 95%	Ratio of successful imports delivered vs. customer deposits taken.
Avg. Delivery Time (Inside Dhaka)	≤ 24 hours	Time from order placement to doorstep for Drop-and-Pack items.
COD Return-to-Origin (RTO) Rate	≤ 8%	Enforced via phone OTP and advance delivery fees on first orders.
Repeat Customer Rate	≥ 20%	Measures customer retention and brand equity.
RMA Turnaround Time (TAT)	≤ 72h (Dhaka) / ≤ 5 days (Nationwide)	Time from claim approval to replacement delivery at customer doorstep.
Dead-on-Arrival (DOA) Rate	< 0.5%	Objective metric of godown QC diligence before parcel handover.
Gross Margin per Drop-and-Pack Order	15%–25% net margin	Evaluated post-runner pickup and packaging overhead.
Supplier Defect Score	Defect rate < 3% per vendor	Wholesale stalls exceeding 3% return rates are suspended.

2. Target Users & Personas

2.1 Persona Matrix

Persona	Description	Core Pain Points	NCAMart Solution
P1: Brand Purist & Quality Seeker	Tech enthusiasts, developers, and corporate professionals buying UGREEN, Baseus, Aqara, Anker.	Prevalent fakes sold as originals; voided distributor warranties.	"Authentic Tier" badge, official distributor warranty, unbroken factory seal, serial number lookup.
P2: Pragmatic Budget Shopper	Students and casual users looking for functional daily cables, adapters, and utility gadgets.	Paying original prices for fakes; devices dead-on-arrival with zero recourse.	"Budget Smart (Tested OEM)" tier with honest labeling, 3-point QC testing, and a 7-day shop replacement warranty.
P3: Field Professional & Technician	Electricians, HVAC technicians, and network engineers needing UNI-T meters or HIKMICRO cameras.	Specialized tools unavailable locally or sold at extreme markups; no formal VAT receipts.	Dedicated Pre-Order catalog for industrial tech, guaranteed genuine procurement, and NBR Mushak 6.3 tax invoices.
P4: Early Tech Adopter	Smart home builders seeking Matter hubs, smart door locks, and portable power stations.	High upfront cost; risky personal imports dead-on-arrival without local support.	Safe pre-order gateway with a 10%–20% deposit, customs clearance pre-screening, and pre-dispatch QC testing.
P5: Operations & QC Staff	The internal fulfillment and testing team at the Nawabpur central godown.	Manual order tracking, mismatched warranty serials, and unorganized vendor returns.	Unified fulfillment dashboard with serial logging, QC checklist, thermal label printing, and runner task management.
P6: Wholesale Merchant Partners	Wholesale stall owners in Sundarban Square, Stadium Market, and Nawabpur Complex.	Fragmented retail orders, delayed cash settlements, and unverified return claims.	Digital runner pickup slips, daily stock status updates, and transparent vendor defect scorecards.
P7: Customer Support & RMA Officers	Internal agents handling customer inquiries, milestone updates, and warranty adjudications.	Disputed customer claims, fraudulent returns, and delayed courier status tracking.	Integrated RMA portal with courier reverse-logistics booking and immutable QC inspection photo archives.

3. Core Features & System Architecture

3.0 Product Categories (Phase 1)

Category	Examples	Default Fulfilment Model
Mobile & Laptop Accessories	Chargers, GaN adapters, cables, power banks, hubs	Drop-and-Pack
Audio & Wearables	Earbuds, headphones, smartwatches	Drop-and-Pack
Small Appliances & Utility Tech	Trimmers, mini fans, extension boards, LED work-lights	Drop-and-Pack
Smart Home & Security	Global-version smart door locks, Matter hubs, robot vacuums, IP cameras	Pre-Order
Professional Tools	HIKMICRO thermal cameras, UNI-T clamp meters & multimeters	Pre-Order
Emergency & Power	Portable power stations, 70mai wireless car jump starters	Pre-Order

3.1 Two-Tier Product Architecture

Every catalog SKU is strictly bound to one of two tiers, clearly marked across search, product cards, comparison tables, and checkout:

1. `Authentic Tier`**:**

- **Visual Badge:** Verified Blue/Gold Authentic Badge.
- **Sourcing:** Authorized channel / official brand distributor.
- **Guarantee:** Official manufacturer warranty + online serial number verification.

1. **Budget Tier ("`Budget Smart (Tested OEM)`"):**

- **Visual Badge:** Teal/Slate Budget Alternative Badge labeled strictly as "`Budget Smart (Tested OEM)`".
- **Sourcing:** Tested unbranded or OEM utility products from vetted local wholesale stalls.
- **Guarantee:** In-house 3-point physical testing + 7-day NCAMart replacement warranty.
- **Strict Policy — Zero Replicas:** Budget Tier must **never** include products that copy another brand's logo, trademark, packaging, or trade dress. Any SKU attempting to mimic an established brand is rejected during catalog onboarding.

1. **Side-by-Side Comparison Module:**

- Programmatically linked between equivalent Authentic and Budget SKUs via `comparison_group_id`.
- Category-specific comparison templates:
- *Chargers/Power:* Output wattage, PD/QC protocol support, heat at full load, safety chips (OVP/OCP/OTP), certifications.
- *Cables:* Max wattage, data speed, connector material, bend-test rating, length.
- *Audio:* Driver size, codec support, battery life, ANC, Bluetooth version.
- *All categories:* Build material, price, warranty coverage, and visual value badges ("Full Safety Suite" vs. "Best Everyday Value").

3.2 Trust & Verification Engine

1. **Cryptographic Base32 Seal ID Entropy:**

- Format: `NCA-` + `4 Alphanumeric (Base32)` + `-` + `4 Alphanumeric (Base32)` (e.g., `NCA-8F2K-9X41`).
- Over 1.09×10^{12} cryptographic combinations generated via `crypto.randomBytes` to prevent sequential enumeration.
- Dual-ID structure: **NCAMart Seal ID** (all items) and **Manufacturer Serial** (`Authentic Tier` items only, bound in database).

1. **Destructive Tamper-Evident VOID Security Seals:**

- Physical labels printed on specialized **Destructive VOID material**. Any attempt to peel or transfer the sticker results in immediate label disintegration and reveals a permanent "VOID" substrate pattern.

1. **Direct Serial & Warranty Lookup Portal (`/verify`):**

- Customers enter a Seal ID or Manufacturer Serial, or scan the QR code printed on the physical seal.
- Returns: Product name, tier badge, QC inspection timestamp, inspecting technician ID, active warranty countdown, and a **"Start Warranty Claim"** button.
- **Rate Limiting & Anti-Scraping:** Protected by Cloudflare Turnstile CAPTCHA and rate limited to max 5 lookups/minute per IP.
- **Cloned Seal Anomaly Detection:** If a single Seal ID is queried from ≥ 10 distinct IP addresses within 24 hours, the portal displays standard info but triggers a high-priority internal alert: `CLONED_SEAL_SUSPECTED`.

1. **Tier-Specific QC Protocols:**

- **Budget Tier / Open-Box — 3-Point QC:** Full unboxing, power/voltage test, port/function check, physical integrity. Technician records checklist in admin, reseals packaging, and applies NCAMart Seal.
- Authentic Tier **, Factory-Sealed — Non-Invasive QC:** Factory seal is **strictly preserved**. Staff inspect brand holograms, match box serials to distributor invoices, and verify packaging integrity. Seal applied over outer protective shrink-wrap.
- **Pre-Order Imports — Comprehensive Pre-Dispatch QC:** Unboxed and functionally tested before domestic dispatch to prevent dead-on-arrival imports.
- **Supervisor 5% Sampling Quota:** System randomly flags 5% of all QC-cleared parcels for mandatory secondary audit by the Godown Manager before courier handover.
- **Printer Jam / Sticker Void Protocol:** If a printer jams or ruins a seal, technician logs a void action with reason (`PRINTER_JAM`, `RIBBON_TEAR`). System marks old ID as `STATUS_VOID_PRINT_ERROR` and generates a fresh Seal ID.

3.3 High-End Pre-Order & Import Engine

1. **Isolated Pre-Order Checkout Flow:**

- **Strict Rule:** Pre-Order items and standard Drop-and-Pack items **cannot be checked out together in the same cart**.
- If a customer attempts to combine items, a modal appears: "*Pre-order items require a separate advance deposit and have different delivery timelines. Please checkout regular items first or clear your cart.*"
- Server-side validation rejects mixed carts with `HTTP 422 Unprocessable Entity` (`ERR_MIXED_CART_DISALLOWED`).

1. **Tiered Deposit Gateway:**

- Advance payment required to lock the order: **10% for items ≥ ৳50,000** and **20% for items < ৳50,000**.
- Paid online via **SSLCommerz** (bKash, Nagad, Cards).

1. **Regulatory Pre-Screening (BTRC & Customs Compliance):**

- Catalog Manager must verify HS Codes and confirm BTRC Type Approval / NOC eligibility for all wireless, RF, and smart-home gadgets prior to catalog publication.

1. **Milestone Tracking:**

- Status progression: *Order Placed (Deposit Verified) → Sourced & In Transit → Arrived at Nawabpur Godown → QC Verified → Out for Delivery (Balance Due via COD/Online)*.

1. **Pre-Order Policies & Default Handling:**

Scenario	Policy & System Action
Import fails / supplier stockout	100% deposit refund within 7 working days, or customer option to switch SKU.
Delay beyond promised ETA + 7 days	Customer may cancel with 100% full refund within 24 hours.
Customer cancels before cargo dispatch	100% deposit refund.
Customer cancels after cargo flight booked	Deposit non-refundable. UI disables cancellation once status reaches `IN_TRANSIT`.
Exchange rate volatility (> 8%)	Landed price includes a 5%–8% currency buffer. If currency devalues > 8%, a Re-Pricing Approval Ticket is sent. Customer may accept adjusted price or receive 100% refund.
Customer default on arrival balance	Automated reminders on Day 1, 4, 7 after QC. On **Day 8**, deposit is forfeited, order cancelled, and item re-listed as an **In-Stock Flash Deal** at landed price.
Batch threshold failure	If N pre-orders not reached within 30 days, system triggers 100% refund or offers individual expedited import fee.

3.4 Local "Drop-and-Pack" Order Management

1. **Proximity-Based Runner Sourcing:**

- High-velocity orders dispatch a digital pickup ticket to the godown runner, who collects the item from Nawabpur, Sundarban, or Stadium partner stalls within 2 hours.

1. **Wholesale Spot Price Tolerance Buffer:**

- If a wholesaler raises counter price, runner inputs updated price into mobile app.
- Automated **5%–7% price tolerance buffer** allows pickup to proceed if net gross margin remains ≥ 15%.
- If price increase exceeds buffer or turns margin negative, runner app freezes task and routes a real-time mobile alert to Sourcing Manager (Approve override, switch to backup stall, or decline & refund).

1. **Offline Runner Sourcing PWA Mode:**

- Mobile web app utilizes Service Workers and IndexedDB. In basement stalls with zero 4G reception, runners can record vendor receipts and prices offline; data auto-syncs upon returning to street level.

1. **Virtual Stock Availability & Fallback:**

- SKUs maintain daily confirmed status (`Available` / `Limited` / `Out`) across primary and backup vendors.
- If all partners are out of stock, customer is notified within 4 hours with options: immediate refund, alternate tier recommendation, or backorder.

3.5 Customer Account & Authentication

- **Phone Number + OTP Login:** Passwordless authentication using Bangladeshi MSISDN validation (`^(?:\+8801|01)[3-9]\d{8}$`).
- **OTP Security Rules:** 6-digit OTP, expires in 3 minutes, max 3 attempts per OTP, max 3 send requests per 10 minutes per phone.
- **Customer Portal:** Order history, pre-order milestone visualizer, **"My Warranties"** (active Seal IDs with countdown timers), saved addresses.
- **Guest Checkout:** Permitted, with mandatory phone OTP verification prior to order placement.

3.6 Payments & Checkout

1. **Primary Payment Gateway — SSLCommerz:**

- Full integration with SSLCommerz handling direct bKash, Nagad, Rocket, Visa, Mastercard, and AmEx transactions.
- PCI-DSS compliant hosted payment flow (no card credentials stored locally).

- **MD5 Hash Callback Verification:** Server validates all IPN payloads using Store Password and server-to-server Order Validation API call.
- **Automated 5-Minute Reconciliation Cron Job:** System queries SSLCommerz Transaction Validation API for any order in `PENDING` state older than 10 minutes, flipping dropped transactions to `PAID` automatically.

1. **Cash on Delivery (COD) & Anti-Fraud Policy:**

- **First-Time Buyers:** Must pay the delivery fee (৳**80 Inside Dhaka / ৳150 Nationwide**) in advance via bKash/Nagad on SSLCommerz to confirm order. Remaining product balance collected on delivery.
- **Verified Repeat Buyers:** Eligible for **100% Cash on Delivery** with zero advance payment required (provided `rto_count == 0` and completed orders ≥ 1).
- **High-Risk Block:** Automated flagging of phone numbers with prior courier return history (`rto_count >= 2` blocks COD access).

1. **Redis 10-Minute Inventory Reservation Lock:**

- To prevent checkout race conditions, starting checkout places a 10-minute temporary Redis reservation on the item. If payment or COD confirmation is not completed in 10 minutes, lock expires.

1. **NBR Mushak 6.3 Digital Tax Invoicing:**

- Checkbox at checkout: "*Request Mushak 6.3 Tax Invoice*". Captures corporate name and BIN.
- Generates an NBR-compliant Mushak 6.3 digital tax invoice PDF with immutable serial logging.

1. **Currency:** All pricing is strictly in **BDT (৳) only**.

3.7 Delivery & Logistics

- **Courier Integration:** Steadfast, Pathao, and RedX with automated API webhook synchronization for parcel tracking.
- **Message Queue & Exponential Backoff:** Courier booking calls decoupled via message queue (BullMQ/RabbitMQ). Retries on failure at 1m, 5m, 15m, 30m before escalating to manual dispatch console.
- **Delivery Zones & SLAs:**
- *Inside Dhaka:* 24–48 hours (Drop-and-Pack items).
- *Outside Dhaka:* 2–5 business days.
- **Showroom Counter Pickup (Nawabpur Complex):**
- Customer receives a 6-digit **Pickup PIN**.
- Showroom staff enters PIN into **Showroom POS Mode** to verify order.
- Customer inspects item on counter test-bench and signs handover slip. Real-time central inventory sync prevents online stockout collisions.

3.8 Returns, Refunds & Warranty (RMA)

1. Budget Smart (Tested OEM) **— 7-Day Replacement:** Covers functional failure within 7 calendar days of delivery. Replaced with identical SKU or refunded if unavailable.
2. Authentic Tier **— Official Manufacturer Warranty:** NCAMart provides an assisted liaison service: collects unit, logs claim with authorized distributor service centers (Star Tech, Ryans, Excel Technologies), and returns repaired unit to customer.
3. **DOA (Dead on Arrival):** Must be reported within 48 hours accompanied by unboxing video. Fast-tracked replacement dispatched within 24 hours.
4. **Reverse Logistics Courier Doorstep Pickup:** Support Agent triggers automated reverse-pickup consignment via courier API. Defective unit collected directly from customer home.
5. **Exclusions:** Physical impact, liquid ingress, severed cables, or tampered/broken NCAMart VOID seals void all warranty coverage.
6. **RMA Turnaround Time (TAT) Target:** Maximum 72 hours for Dhaka; maximum 5 business days nationwide.

3.9 Role-Based Access Control (RBAC) Matrix

System Resource / Action	Customer	Field Runner	QC Tech	Showroom POS Staff	Sourcing Manager	Support Agent	Super Admin
Browse Catalog & Verify Seal	✅ Read	✅ Read	✅ Read	✅ Read	✅ Read	✅ Read	✅ Full
Place Order / Pay Deposit	✅ Create	❌	❌	❌	❌	❌	✅ Full
View Runner Pickup Tasks	❌	✅ Read/Update	❌	❌	✅ Read/Assign	❌	✅ Full
Input Wholesale Purchase Price	❌	✅ Input	❌	❌	✅ Override	❌	✅ Full
Execute QC Checklist & Print Seal	❌	❌	✅ Execute	❌	❌	❌	✅ Full
Void Damaged Seal ID	❌	❌	❌	❌	✅ Approve	❌	✅ Full
Showroom POS Counter Checkout	❌	❌	❌	✅ Execute	❌	❌	✅ Full
View Customer PII (Full Details)	Own profile	❌ (Masked)	❌ (Masked)	Counter only	❌ (Masked)	✅ Read	✅ Full
Approve Spot Price Over-Buffer	❌	❌	❌	❌	✅ Approve	❌	✅ Full
Adjudicate RMA Warranty Claim	Own claims	❌	❌	❌	❌	✅ Process	✅ Full
Issue Financial Refund	❌	❌	❌	❌	❌	Up to ৳1,000	✅ Full
Edit Catalog / Product Tiers	❌	❌	❌	❌	❌	❌	✅ Full
View Financial & Profit Reports	❌	❌	❌	❌	❌	❌	✅ Full

Security Rule on PII Masking: Field runners only see Order #, Item SKU, Wholesale Stall #, and Max Purchase Price. Full shipping addresses are restricted to the central godown packaging station.

3.10 Multi-Channel Notification Matrix

Lifecycle Trigger	SMS	WhatsApp	Email	Delivery SLA / Window	Quiet-Hour Policy
1. OTP Verification Code	✅	—	—	Real-time (< 15 sec)	24/7 (Exempt)
2. Order Placed (Pre/Drop)	✅	✅	✅	Instant (< 1 min)	24/7 (Exempt)
3. Advance Delivery Fee Confirmed	✅	✅	✅	Instant (< 1 min)	24/7 (Exempt)
4. Pre-Order Sourced / In Transit	✅	✅	✅	Within 2 hours of update	08:00 AM – 10:00 PM
5. Pre-Order QC Passed / Balance Due	✅	✅	✅	Instant upon QC pass	08:00 AM – 10:00 PM
6. Out for Delivery Alert	✅	✅	—	Morning of delivery	08:00 AM – 10:00 PM
7. Pre-Order Balance Reminder (D1,4,7)	✅	✅	✅	Morning (10:00 AM)	08:00 AM – 10:00 PM
8. RMA Status Change	✅	✅	✅	Within 1 hour of review	08:00 AM – 10:00 PM
9. Cloned Seal Alert (Security)	Admin	Admin	Admin	Urgent Real-time (< 1 min)	24/7 (Internal)

Audit log of every staff action that changes data.

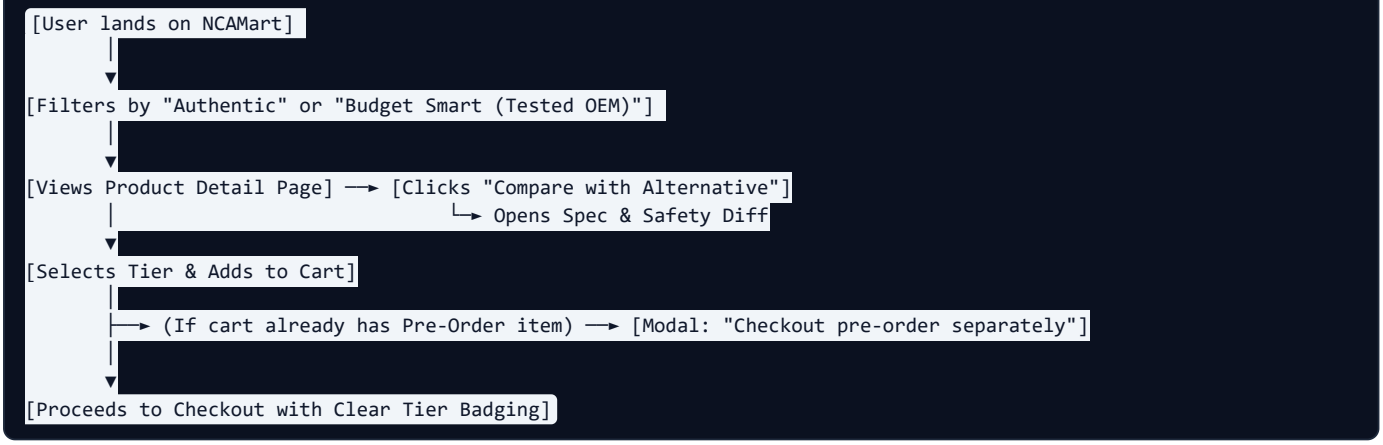
BTRC Compliance: In compliance with BTRC regulations, non-transactional messages are restricted between 10:00 PM and 08:00 AM. System incorporates circuit breaker logic falling back from SMS to WhatsApp on delivery failure.

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4. End-to-End User & Operational Flows

Flow 1: Transparent Browsing, Comparison & Isolated Cart

OPERATIONAL WORKFLOW DIAGRAM



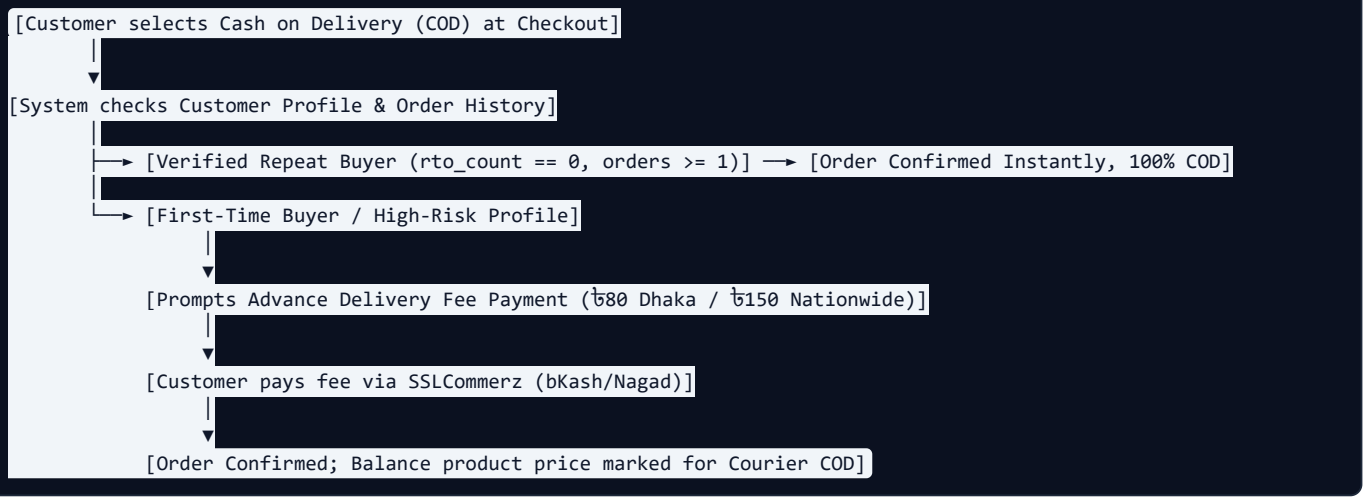
Flow 2: Pre-Ordering Niche Tech & Milestone Tracking

OPERATIONAL WORKFLOW DIAGRAM



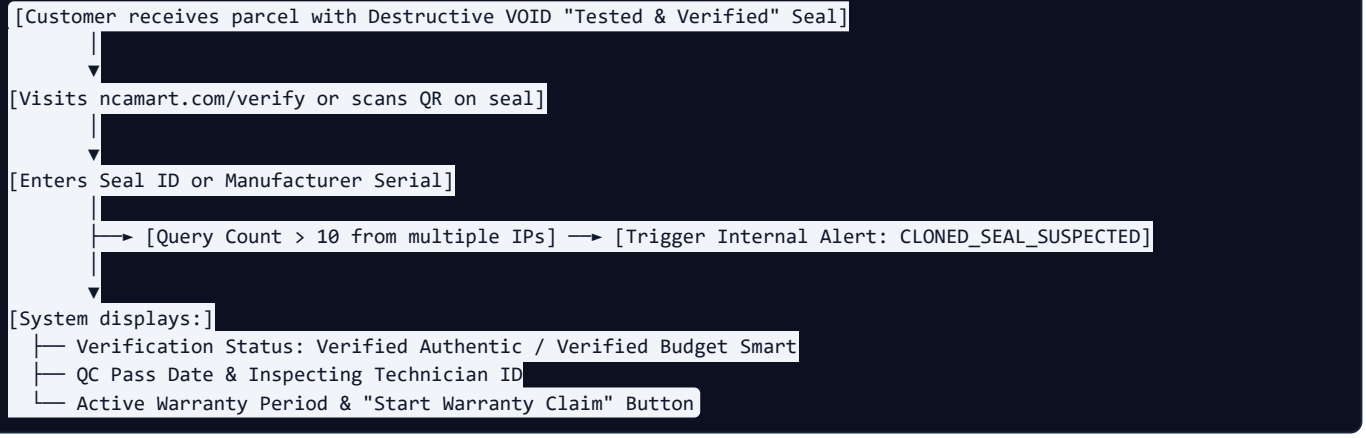
Flow 3: COD Checkout with First-Time Buyer Delivery Fee

OPERATIONAL WORKFLOW DIAGRAM



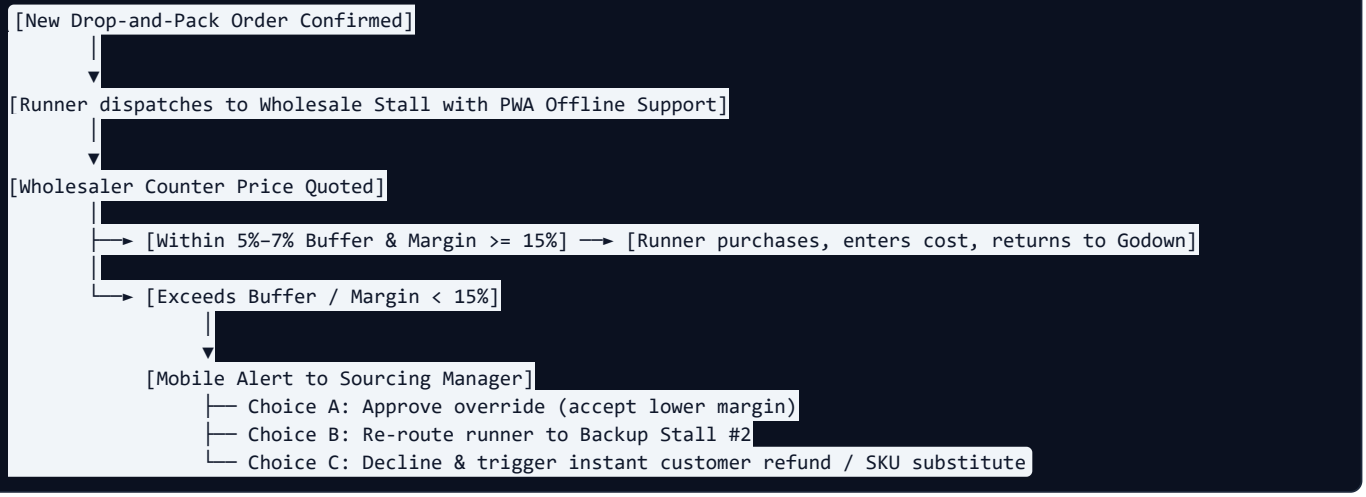
Flow 4: Post-Purchase Serial Verification & Cloned Seal Check

OPERATIONAL WORKFLOW DIAGRAM



Flow 5: Drop-and-Pack Sourcing & Spot Price Tolerance Workflow

OPERATIONAL WORKFLOW DIAGRAM



Flow 6: Showroom Counter Pickup & Inspection (Nawabpur Complex)

OPERATIONAL WORKFLOW DIAGRAM

```
[Customer selects "Showroom Counter Pickup" at Checkout]
       │
       ▼
[Godown stages item in Counter Locker + generates 6-digit Pickup PIN via SMS]
       │
       ▼
[Customer arrives at Nawabpur showroom & provides PIN]
       │
       ▼
[Showroom Staff enters PIN into Admin POS Mode]
       │
       ▼
[Customer tests device on counter test-bench & signs handover]
       │
       ▼
[Order marked DELIVERED_COUNTER; Real-time inventory updated]
```

Flow 7: Failed Delivery & Return-to-Origin (RTO) Lifecycle

OPERATIONAL WORKFLOW DIAGRAM

```
[Courier attempts delivery 3 times / Customer refuses parcel]
       │
       ▼
[Courier API webhook: DELIVERY_FAILED_RTO_INITIATED]
       │
       ▼
[Customer record flagged: rto_count += 1] ──(rto_count >= 2)──► [Block future COD access]
       │
       ▼
[Parcel returns to Nawabpur Godown (3–5 business days)]
       │
       ▼
[QC Technician inspects parcel exterior]
       ├── [Seal Intact] ──► Restock in Drop-and-Pack inventory or return to vendor
       └── [Seal Damaged/Crushed] ──► Lodge transit damage claim against courier
```

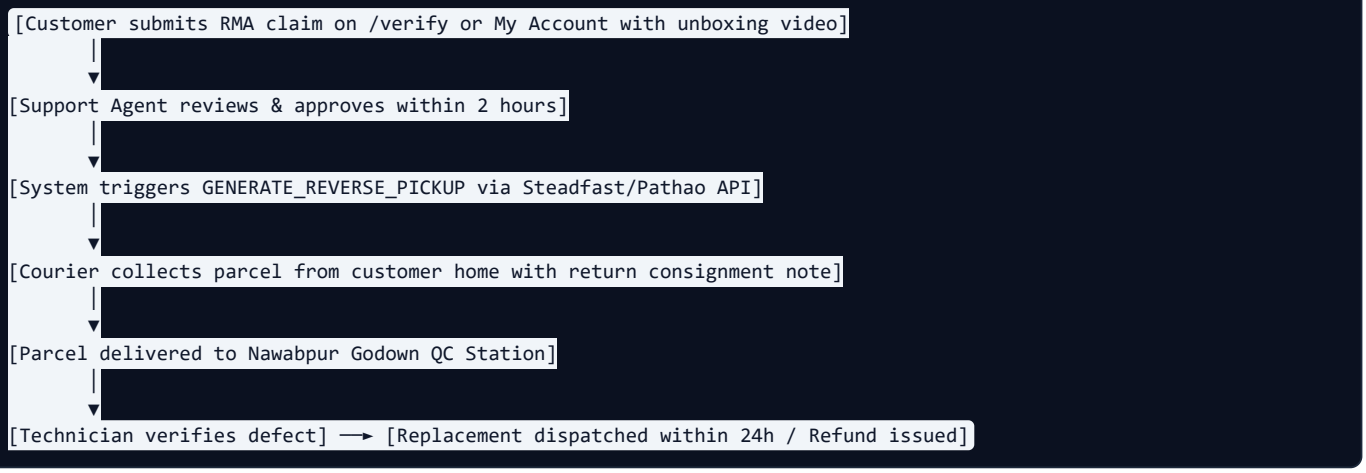
Flow 8: Pre-Order Customer Default & Inventory Forfeiture

OPERATIONAL WORKFLOW DIAGRAM

```
[Pre-Order Item passes QC at Nawabpur Godown]
       │
       ▼
[Day 1: Automated SMS/WhatsApp: Balance due notification]
       │
       ▼ (No payment)
[Day 4: Warning notice: 72 hours remaining]
       │
       ▼ (No payment)
[Day 7: Support agent manual verification phone call]
       │
       ▼ (No response / Refusal)
[Day 8: System triggers DEPOSIT_FORFEITED_CANCELLED]
       │
       ▼
[Deposit forfeited per Terms; Item auto-listed as "In-Stock Flash Deal" at landed price]
```

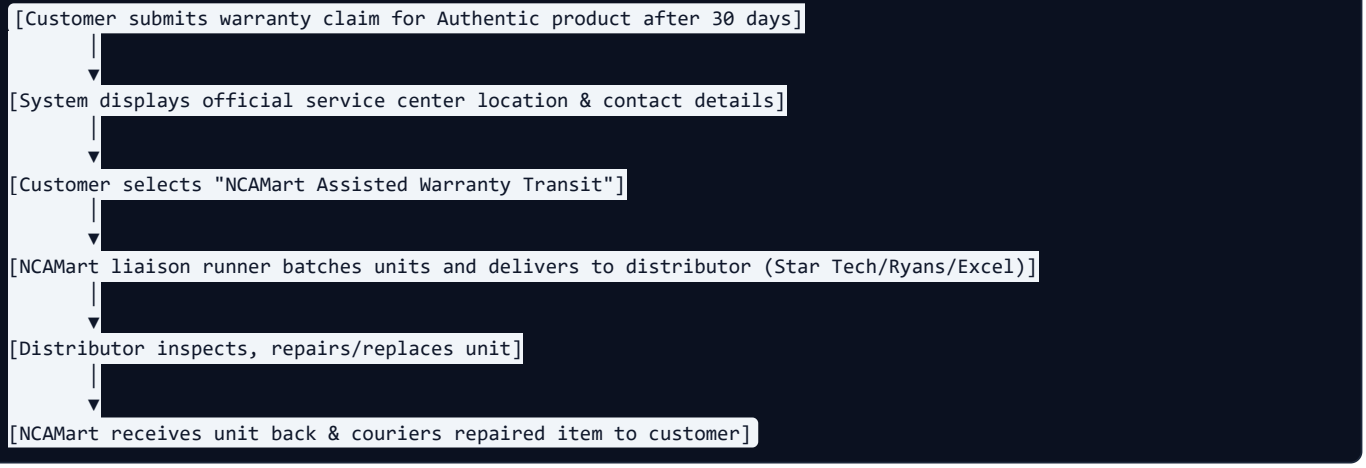
Flow 9: Reverse Logistics (Courier Doorstep RMA Pickup)

OPERATIONAL WORKFLOW DIAGRAM



Flow 10: Authentic Brand Distributor Warranty Escalation

OPERATIONAL WORKFLOW DIAGRAM



5. Out of Scope (Phase 1 / MVP)

1. **Open Multi-Vendor Marketplace:** Third-party sellers cannot list items. All stock is curated and QC-inspected by NCAMart.
2. **Automated International Dropshipping:** Direct shipping from overseas suppliers to end consumers without Nawabpur physical QC is strictly prohibited.
3. **Complex B2B Credit Financing:** Wholesale line-of-credit or net-30 billing for contractors is reserved for Phase 2.
4. **Custom Hardware POS Terminals:** Showroom counter sales use the responsive web-based admin portal rather than dedicated POS hardware.
5. **Bank EMI Gateways:** Institutional installment plans deferred to Phase 2; Phase 1 uses the tiered deposit model.
6. **Native Mobile Apps (Android/iOS):** Mobile-first responsive web application (PWA-ready). Native apps reserved for Phase 2.
7. **Unified Mixed-Cart Splitting Engine:** MVP enforces separate checkouts for Pre-Order and regular items to eliminate logistics friction.

6. Non-Functional & Operational Requirements

- **Performance & Mobile Responsiveness:** Mobile-first architecture optimized for Bangladesh mobile 4G/3G networks. Targets: FCP < 1.2s, LCP < 2.5s on 4G, Lighthouse mobile score ≥ 85.
- **Security & Anti-Fraud Architecture:**
- SSLCommerz hosted checkout redirect; zero local card credential storage.
- Cryptographically randomized Base32 Seal IDs (`NCA-XXXX-XXXX`) to eliminate sequential ID enumeration.
- Rate limiting on OTP, login, and `/verify` endpoints.
- Sensitive customer PII masking on runner mobile interfaces.
- 5% supervisor random inspection quota on sealed packages.
- **Data Integrity & Traceability:** Immutable database logging for Seal IDs, QC test results, serial bindings, and warranty timestamps.
- **Localization:** Bilingual UI (Bangla & English). **All pricing in BDT (৳) only.**
- **Reliability:** 99.5% uptime SLA. Daily automated database backups with 30-day retention and point-in-time recovery (PITR).
- **SEO & Discovery:** Server-side rendered (SSR) product pages with structured JSON-LD Schema (`Product` , `Offer` , `Brand` , `Breadcrumbs`).

7. Data Model (High-Level Schema Entities)

Entity	Key Fields
Product	id, name_bn, name_en, category, tier (`authentic` / `budget_smart`), fulfilment_model (`drop_pack` / `pre_order`), comparison_group_id, btrc_cleared (bool)
Variant / SKU	id, product_id, sku_code, price_bdt, cost_price_bdt, deposit_pct, spot_buffer_pct
Supplier	id, name, market (Nawabpur/Sundarban/Stadium/Patuatuli/Import), stall_no, phone, defect_score
SupplierStock	supplier_id, sku_id, status (`available` / `limited` / `out`), updated_at
Order	id, order_number, customer_id, items, payment_method, payment_status, delivery_zone, advance_fee_paid, courier, tracking_no, status, mushak_requested (bool), pickup_pin (nullable)
PreOrder	id, order_id, deposit_bdt, balance_bdt, milestone, eta_date, price_locked_bdt, forfeiture_date
SealRecord	seal_id (alphanumeric Base32), sku_id, manufacturer_serial (nullable), order_id, technician_id, warranty_start, warranty_end, status (`active` / `claimed` / `void`), void_reason
QCRecord	id, seal_id, qc_type (`3_point` / `non_invasive` / `full`), test_checklist (json), result (`pass` / `fail`), technician_id, supervisor_checked (bool), created_at
WarrantyClaim (RMA)	id, claim_number, seal_id, customer_id, issue_description, media_urls, status, resolution (`replace` / `repair` / `refund`), reverse_consignment_no
Customer	id, phone (verified), name, email, addresses, order_count, rto_count, cod_blocked (bool)
Staff	id, name, role (`super_admin` / `catalog` / `sourcing` / `runner` / `qc` / `pos` / `support`), 2fa_enabled
AuditLog	id, staff_id, action, entity, before_state, after_state, timestamp

8. Legal & Regulatory Compliance

1. **Trademarks Act, 2009 Compliance:** Zero tolerance for replicas or counterfeit brand names. Budget Tier SKUs are restricted to unbranded or OEM utility hardware.
2. **Digital Commerce Operation Guidelines (Bangladesh):** Adherence to national rules regarding delivery timelines, refund SLA (7 days), truthful disclosure, and escrow compliance for advance payments.
3. **BTRC Regulatory Compliance:** Pre-screening of all imported wireless/RF devices to confirm Type Approval or NOC eligibility before offering them in the pre-order catalog.
4. **NBR Tax Compliance:** Capability to issue National Board of Revenue (NBR) compliant Mushak 6.3 digital tax invoices for B2B/contractor sales.
5. **Business Licensing:** Active Trade License, e-TIN, BIN (VAT registration), and DBID (Digital Business Identification) with the Ministry of Commerce.
6. **Mandatory Storefront Policies:** Terms & Conditions, Privacy Policy, Return & Refund Policy, Warranty & RMA Terms, Pre-Order Policy, and clear physical location disclosures.

9. Marketing & Sales Channels

- **Facebook & Instagram Shop:** Automated catalog sync with direct WhatsApp/Messenger shopping assistance.
- **WhatsApp Business Automation:** Instant order confirmation, pre-order milestone tracking, and customer service.
- **Content Marketing & SEO:** Educational comparison content (e.g., "*How to spot a fake GaN charger in Dhaka*"), establishing NCAMart as the authority on tech authenticity.
- **Google Merchant Center:** Structured product feed for Google Shopping queries.
- **Physical Walk-In QR Integration:** Standees at Nawabpur Complex showroom driving traffic to `/verify` and online pre-orders.

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10. Risks, Mitigations & Exceptions Matrix

Risk / Exception Factor	Severity	Strategic Mitigation & Escalation Protocol
Wholesale spot price increase	Medium	5%–7% price tolerance buffer; automated approval escalation to Sourcing Manager (§3.4).
Import / customs clearance delay	High	Pre-screen BTRC NOC requirements (§3.3); conservative ETAs (15–25 days); proactive milestone alerts. If delay > 7 days, offer ৳1,000 store voucher or instant 100% refund.
USD/BDT exchange rate spikes (> 8%)	High	Landed price includes 5%–8% currency buffer. Re-Pricing Approval Ticket issued; customer may accept adjusted price or cancel with 100% refund.
High COD return-to-origin (RTO)	High	Mandatory phone OTP; advance delivery fee (৳80/৳150 via SSLCommerz) for first-time buyers; `rto_count >= 2` blocks COD.
Seal peeling & warranty fraud	High	Destructive VOID security stickers (§3.2); immutable QC dispatch logging; Cloned seal anomaly alerts.
Mixed-order fulfillment conflicts	Medium	Separate checkout flows for Pre-Order vs. Drop-and-Pack items (`ERR_MIXED_CART_DISALLOWED`).
Showroom vs. Online stock collision	Medium	Real-time Showroom POS counter mode in admin portal with instant inventory lock.
Supplier batch defect failure	Medium	If 3 of 5 units fail QC, entire batch is returned for cash/exchange; supplier defect score increases; stalls > 3% defect rate are delisted.
Courier loss or transit damage	Medium	Immediate replacement dispatched to customer within 24 hours of unboxing video; NCAMart handles courier insurance claim internally.
Distributor warranty refusal	High	NCAMart 100% Authentic Guarantee SLA activates: NCAMart assumes full financial liability, issuing new replacement or refund within 48h.

11. Release Roadmap & Action Plan

Phase	Milestone Scope	Target Timeline	Priority Focus
Phase 0 — Foundation	Business licensing (DBID/BIN), supplier vendor onboarding, thermal VOID seal sourcing, brand design system	Weeks 1–3	**P0** Security & Legal
Phase 1 — MVP Launch	Two-tier catalog, comparison module, SSLCommerz gateway (bKash/Cards/COD), separate checkout flows, `/verify` portal, admin godown QC & thermal label printing, courier API, showroom POS mode, RTO workflow	Weeks 4–10	**P0 & P1** Core Operations
Phase 1.5 — Pre-Order Engine	Milestone tracking, BTRC pre-screened import catalog, deposit engine, 7-day default forfeiture workflow, WhatsApp automation	Weeks 11–14	**P1 & P2** Scaling Trust
Phase 2 — Growth & Scaling	Bank EMI integrations, native mobile apps (Android/iOS), B2B contractor credit accounts, private-label branding	Month 4+	**P3** Expansion

12. Architectural & Operational Decisions (Resolved)

1. **Pre-Order Deposit Structure: Tiered by Price** — **10% advance deposit** for high-ticket items (≥ ৳50,000), **20% advance deposit** for items < ৳50,000.
2. **Primary Payment Gateway: SSLCommerz** (integrating direct bKash, Nagad, Rocket, Visa, Mastercard, and AmEx).
3. **Budget Tier Branding:** Initially labeled purely as "`Budget Smart (Tested OEM)`". No replicas permitted. (Private label deferred to Phase 2).
4. **Exchange Rate Fluctuation Threshold:** Landed pricing includes a built-in currency buffer (5%–8%); if exchange rate devaluation exceeds 8%, price is updated with customer approval or a 100% refund is issued.
5. `Authentic Tier` **Extended Warranty: No.** Authentic items rely strictly on official manufacturer/distributor warranties; no shop add-on warranty will be sold.
6. **Tech Stack & Infrastructure: Next.js (App Router, SSR/PWA) + PostgreSQL + Headless Admin**, deployed on a low-latency local or Singapore-region (SG) VPS.